

Hospital Management System (HMS) - Application Overview

This document provides a comprehensive overview of the Hospital Management System (HMS), a high-performance, full-stack application designed for clinical operations, patient management, and hospital logistics.

Technical Architecture

The HMS is built using a modern decoupled architecture, ensuring scalability and performance.

****Frontend****

- ****Framework:**** React 19 (via Vite)
- ****Styling:**** Vanilla CSS (Custom Design System)
- ****State Management:**** React Hooks & Context API
- ****Routing:**** React Router v7

****Backend API****

- ****Runtime:**** Python 3.10+
 - ****Framework:**** FastAPI
 - ****Authentication:**** Custom `X-User-Id` Header-based Authentication
 - ****Database:**** SQLite 3 (Persistent storage via `hms.db`)
 - ****Dependencies:**** SQLAlchemy, Pydantic, Uvicorn
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Database Schema

The system utilizes a relational SQLite database with the following core entities:

Table	Description
`users`	System administrators and staff login credentials.
`patients`	Comprehensive registry of patient profiles and medical status.
`appointments`	Scheduling records for patient-doctor consultations.
`records`	Clinical histories, diagnoses, and prescriptions.
`invoices`	Billing and revenue tracking data.
`medicines`	Pharmacy inventory and stock management.
`staff`	Human Capital management (Doctors, Nurses, Admin).

``blood_inventory``

Real-time tracking of blood groups and units.

``inventory``

General hospital logistics and supply tracking.

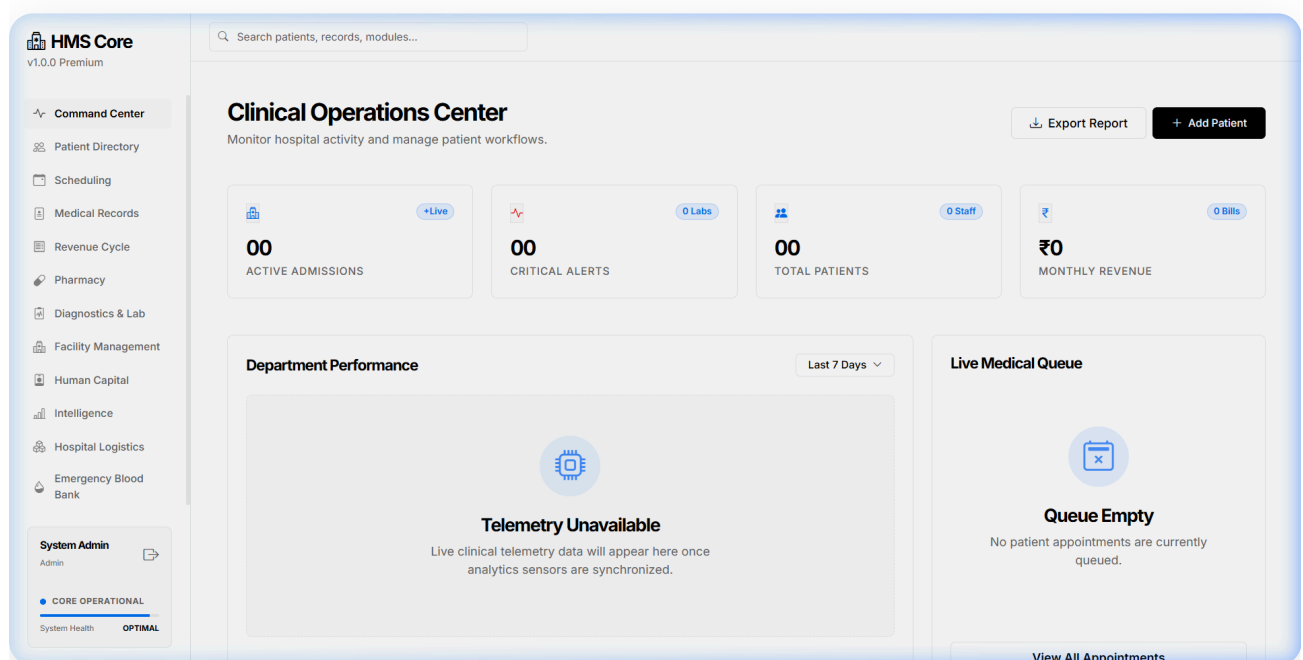


Visual Interface Gallery

Below are live captures of the HMS Core interface, highlighting the premium design and operational dashboards.

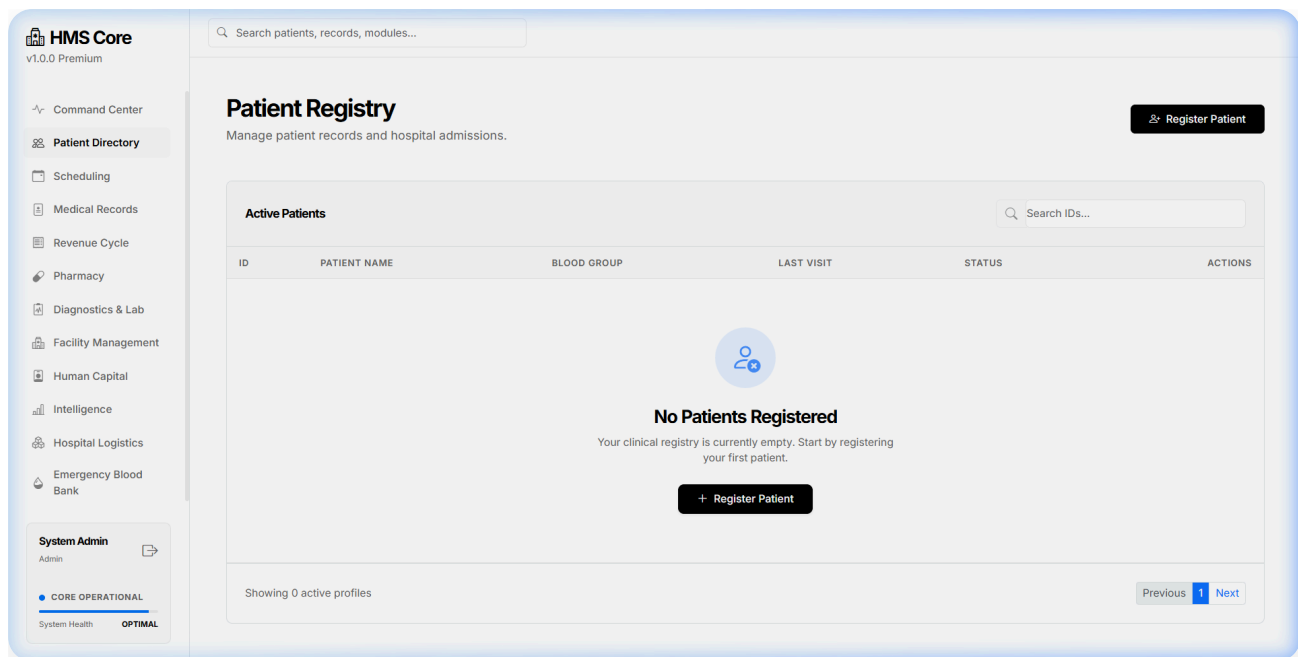
****1. Command Center (Dashboard)****

The central hub for clinical telemetry, active admissions, and revenue metrics.



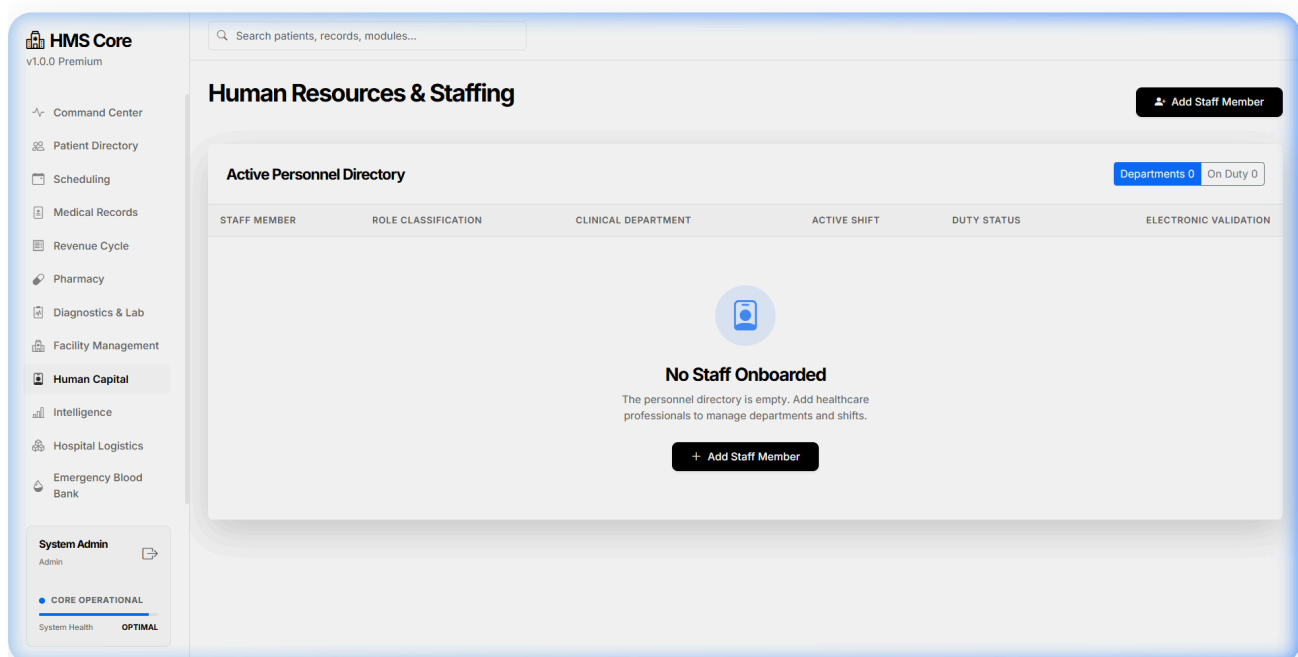
****2. Patient Registry****

Secure management of electronic health records and admission workflows.



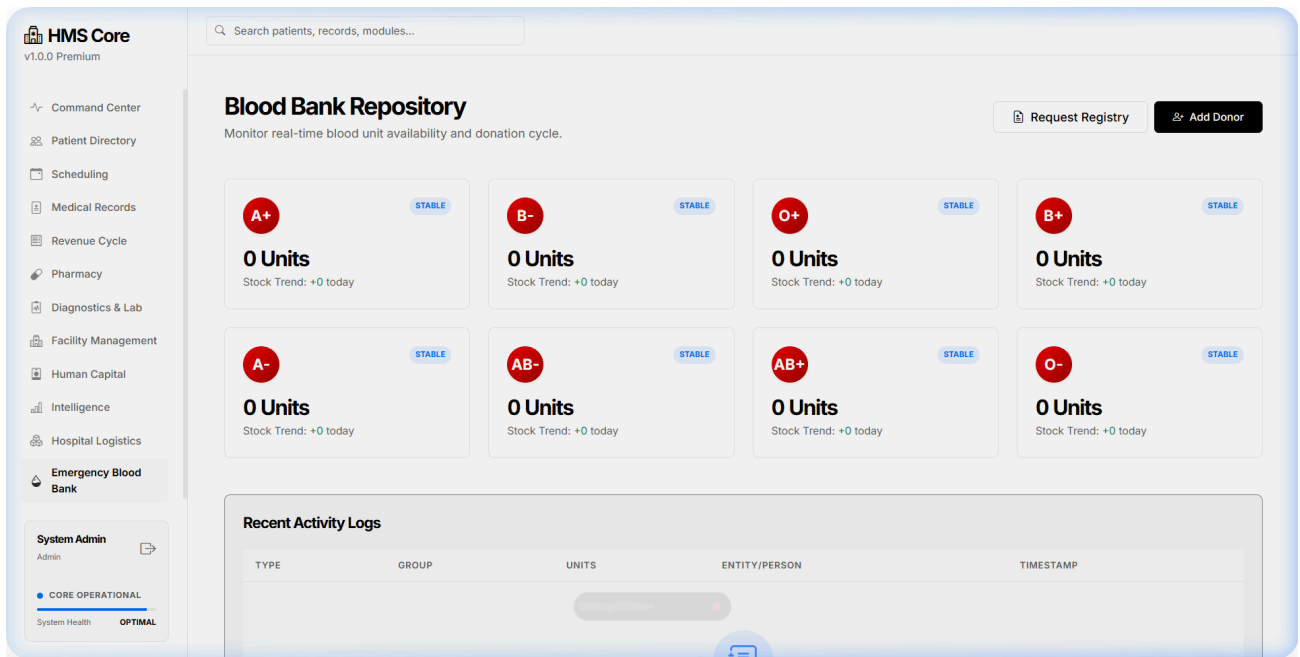
****3. Human Capital (Staff)****

Comprehensive directory of hospital departments and active shifts.



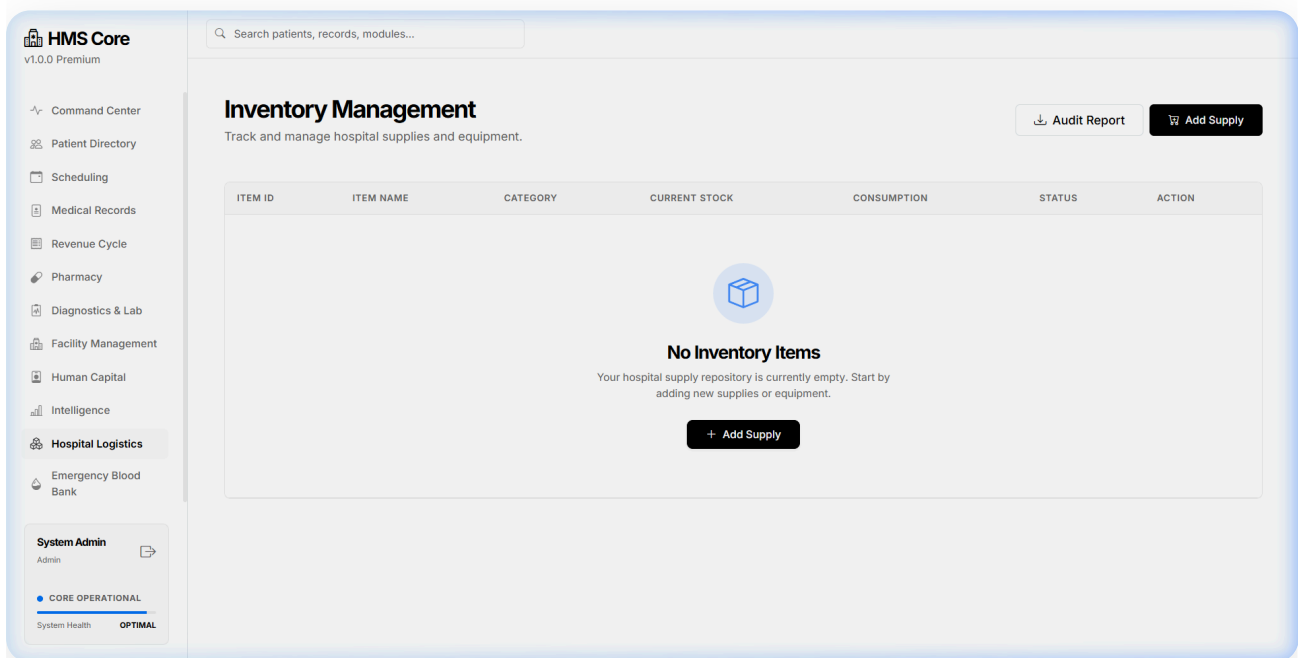
****4. Emergency Blood Bank****

Real-time monitoring of blood units and supply trends.



****5. Hospital Logistics (Inventory)****

Automated supply chain tracking for clinical and general hospital items.



Execution & Deployment

****Frontend Development****

```
npm run dev
```

Starts the Vite development server on `http://localhost:5173`.

****Backend Development****

```
cd backend  
python -m uvicorn app.main:app --reload --port 8000
```

Runs the FastAPI server on `http://localhost:8000`.

Note: This document was generated automatically by Antigravity based on a real-time audit of the codebase and UI.